

Dynamic Trajectory Tracking by Synergistic Dual-Stage Actuation and Control

Cheng-Wei Chen , Yen-Chi Chang, and Tsu-Chin Tsao 

Abstract—This paper addresses precision tracking control of dual-stage actuators for machining of noncircular engine piston profiles. To achieve desired tracking performance, the capacity limits, dynamic characteristics, external disturbances, and model uncertainties of the dual-stage actuators pose significant challenges in control design and real-time implementation. The master–slave linear control configuration previously applied to similar applications is inadequate when nonlinear dynamics are triggered by constraint violations. Optimal predictive control, which minimizes the tracking error subject to physical constraints, is desirable, but the substantial computation forbids real-time implementation at the sampling rate required by the application. A hybrid approach is proposed where an offline constrained optimal feedforward tracking utilizing the model-predictive control framework (FFMPC) is added to a two-parameter robust repetitive control (TPRRC) to achieve high-precision tracking performance, which would otherwise be unattainable solely by each individual control action. Multirate digital control implementation on a dual-stage actuator system, which is comprised of a long-stroke low-bandwidth linear motor and a short-stroke high-bandwidth piezoelectric actuator for tracking noncircular engine piston profiles is presented. The experimental results demonstrate the effectiveness of the approach, where the tracking error by TPRRC is reduced by 84.8% when FFMPC is added.

Index Terms—Linear synchronous motors, machine tool control, mechatronics, motion control, piezoelectric devices.

I. INTRODUCTION

DUAL-STAGE actuation refers to single-axis mechanical motion generated by the cascaded motion of two actuators. The primary actuated stage is with coarse precision, large stroke, and low bandwidth. The secondary actuated stage is with finer precision, shorter stroke, and higher bandwidth. Dual-stage actuation is used in challenging dynamic motion and vibration compensation, where a single actuator cannot satisfy the performance requirement. Examples of dual-stage actuation include

computer hard disk drive read–write heads [1], [2], noncircular turning process for engine cam lobes machining [3], raster scan cyclic motions [4], [5], and long-stroke fine-precision positioning [6], [7]. This work is motivated by the application of noncircular turning of engine pistons, where some profiles have complex geometry significantly deviated from the typical oval cross-sectional shapes. The requirement of machining complex profiles justifies the use of the dual-stage actuation method.

Dual-stage actuation implies that each actuator has its own strength and limitation so that the main goal for the controllers is to minimize the output error by generating the control signals to the actuators subject to hardware limitations [1]–[7]. Without taking the hardware limitation into consideration explicitly, the high-frequency components can be extracted heuristically from the reference input and assigned to the fast-responding secondary stage [1]. An effective extension of this heuristic approach is the master–slave configuration [6], [7], which assigns the output error of the primary-stage control loop as the reference command to the secondary-stage control loop. The primary-stage control loop rejects low frequencies of the input, and its error, which becomes the high-pass filtered input, is sent to the secondary-stage control loop for rejecting higher frequency components.

High-gain feedback control based on the internal model principle [8] for specific frequencies of interest and optimal feedforward control based on approximated dynamic inversion of nonminimum phase systems [9] are often employed in high-performance precision motion control. A wealth of literature has documented the success of employing either or both to achieve high-performance tracking control [1], [10]–[16]. Among these work, the two-parameter robust repetitive controller (TPRRC) [10] and its extension with additional feedforward control action [17] formulated and solved as a μ -synthesis problem [11] have been demonstrated to precisely track periodic or near-periodic signals in applications such as noncircular turning of engine camshaft lobes [10], [17] and piston profiles [18]. The TPRRC has also been applied to the master–slave configuration of dual-stage electrohydraulic and piezoelectric actuators for tracking engine cam profiles. It demonstrated far superior tracking performance with 75% reduction in tracking errors when compared to the single-stage electrohydraulic actuator alone [3].

The aforementioned TPRRC or linear filtering approach does not explicitly address the hardware limits or constraints. Obviously, the high-gain feedback and dynamic inversion feedforward are particularly susceptible to saturation violations. Dynamic inversion in the low-gain frequency region

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creates high feedforward gain that may cause input saturation. Integrator windup and input saturation induced instability, and their treatments have been documented extensively [19]–[23]. Starting with the high-performance controller like TPRRC, this paper proposed an add-on solution to address the issue of deteriorating performance that may be caused by violating the hardware limits.

Optimal control subject to constraints and uncertainties of the plant model and external disturbances is a salient feature of the receding horizon model-predictive control (MPC) [24], [25], which seems to be ideal to meet the need of the present dual-stage actuation control problem [26], [27]. Ensuring MPC stability requires prudent selection of parameters such as receding horizon length, terminal conditions, and modification of the cost function from the original intention [25]. While MPC has been demonstrated by simulation [26] and experiment of sampling rate lower than hundreds of cycles per second for mechanical motion control [27], real-time implementations of MPC for dynamic motion control, which customarily has a sampling rate higher than 2 kHz, have been scarce except for a few cases such as a single actuator with low-order (first- or second-order) dynamics [12], or a multi-axis system with highly exploited special optimization reduction [28]. Real-time optimization for MPC, though an active research area with significant advances made [29], is infeasible for this application due to its fast sampling rate.

To create a solution feasible for real-time implementation, we first design the master–slave TPRRC feedback controllers as if there were no hardware limits or constraints. With the closed-loop control structure, we then perform an offline constrained receding horizon optimization [30], [31], termed as feedforward model predictive control (FFMPC) by abuse of terminology, to create the optimal feedforward inputs to the two actuators. Since this is an offline receding horizon optimization for the feedforward action to a stable closed-loop system, we are free to choose sufficiently large receding horizon so that the requirements of terminal conditions or control effort to ensure stability can be relaxed. As such, the performance index represents exactly the intention of the performance metric. Thus, the FFMPC minimizes the tracking error subject to the constraints, while the TPRRC feedback control reduces the effect of uncertainties from the modeling error and disturbances. Adequate head rooms may be included in the FFMPC constraints to account for uncertainties in models and disturbances. For the noncircular turning process application, the disturbance force is typically a small fraction of the force required to generate the dynamic motion.

Compared to the preliminary simulation of the proposed approach [32], which has shown promising results, this paper provides comprehensive experimental validation and discussion. The main contributions of this paper are as follows:

- 1) combining robust repetitive control with constrained feedforward MPC for precise tracking while satisfying constraints;
- 2) formulating and solving a quadratic programming (QP) for the optimal feedforward inputs to the dual-stage master–slave TPRRC configuration;

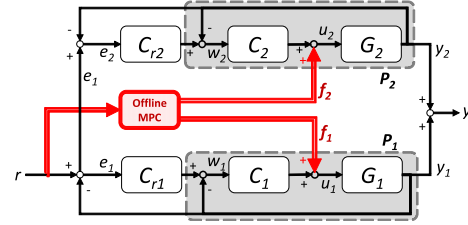


Fig. 1. Block diagram of the FFMPC applied to the dual-stage TPRRC.

- 3) implementing the TPRRC and FFMPC on a multirate dual-stage system, consisting of a linear motor (LM) and a piezoelectric actuator (PZT), and demonstrating superior performance for tracking an engine piston profile.

The rest of this paper is organized as follows. Section II shows the master–slave configuration of the dual-stage control. Section III illustrates the QP formulation of the constrained FFMPC with the consideration of hardware limits. The performance is demonstrated by simulation and experimental results on a multirate dual-stage system in Section IV. Concluding remarks are presented in Section V.

II. DUAL-STAGE FAST TOOL SERVO SYSTEM

The dual-stage fast tool servo system was developed to handle the challenging dynamic references, which contain long traveling distance and high-frequency components. In general, the primary stage G_1 handles the low-frequency movement, and the secondary stage G_2 compensates for the high-frequency components in the trajectory by utilizing its fast-responding capability. The overall displacement y is the summation of both stages' displacement, y_1 and y_2 .

As shown in Fig. 1, the master–slave control configuration is incorporated with the TPRRC to address the near-periodic components coupled to the work piece rotation in the CNC machining process [3]. Compared to the parallel scheme that has been applied in many applications [1], the master–slave configuration takes advantage of its lower sensitivity, which is the multiplication of individual sensitivity function of each stage. In other words, the primary stage serves as a low-pass filter, and the error e_1 caused by high-frequency components in the trajectory is submitted to the secondary fast-responding stage. The resulting tracking error e_2 for the secondary stage equals the total tracking error, $r - y$.

Though not required, actuators G_1 and G_2 may be stabilized by simple feedback controllers, namely, C_1 and C_2 , for the robust repetitive controller design

$$P_i = \frac{C_i G_i}{1 + C_i G_i}, \quad i = 1, 2 \quad (1)$$

where P_i are the closed-loop plants. In the TPRRC algorithm [10], the repetitive controllers C_{ri} are obtained by μ -synthesis [33] based on the plant models P_i and properly selected weighting functions that characterize performance and uncertainties. Fig. 2 shows the setup of the μ -synthesis problem on a single stage, where K_1 stabilizes the positive feedback loop and K_2 improves performance. The key is that the generally long delay

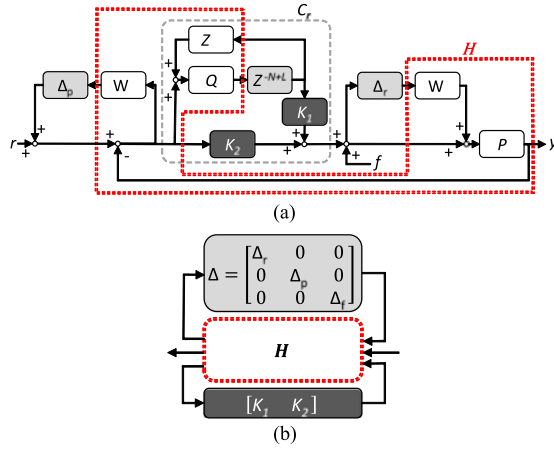


Fig. 2. Block diagram of the TPRRC. (a) Framework of the TPRRC. (b) LFT form of the TPRRC.

Z^{-N+L} in the repetitive control's internal model has been replaced by a fictitious complex uncertainty Δ_f to significantly reduce the order of the augmented plant model. μ -synthesis that solves for K_1 and K_2 such that $\mu_\Delta(F_l(H, [K_1, K_2])) < 1$ implies the robust stability $|W_r T| < 1$ and nominal performance $|W_p S| < 1$. Note that H is the generalized plant as illustrated in Fig. 2, $F_l(\cdot, \cdot)$ denotes lower linear fractional transformation (LFT), W_r is an uncertainty weighting function, W_p is the sensitivity performance weight, and S is the sensitivity function

$$S = 1 - T = \frac{1}{1 + C_r P} \quad (2)$$

where T is defined as the complementary sensitivity.

In the TPRRC design, the sensitivity function S approximates zero at the fundamental frequency and its harmonics. In the case that the desired trajectory is purely periodic, the combination of the dual-stage system and repetitive controllers is able to achieve high-performance trajectory tracking. However, in reality, perfect oval shape is rarely adopted. To overcome the long transient of learning caused by robust repetitive controllers, model-based feedforward signals are necessary to be cascaded on the feedback control loops. Most importantly, the physical hardware limits of the actuators also need to be considered for avoiding constraint violations, which happen when using TPRRC controllers to track aggressive dynamic references.

III. OPTIMIZATION FORMULATION OF CONSTRAINED FEEDFORWARD WITH ROBUST REPETITIVE CONTROL

In order to prevent constraint violations and improve the tracking performance of the dual-stage system at the same time, a constrained FFMPC is combined with the TPRRC. Fig. 1 shows the block diagram of the proposed combination of FFMPC and TPRRC for dual-stage tracking. The reference signal is r , and the output y is the summation of both stages y_1 and y_2 . The feedforward signals, f_1 and f_2 , are offline calculated after solving the optimization and directly fed to the actuators, G_1 and G_2 . By cascading the feedforward signals to the feedback actions from the TPRRC, this configuration can achieve high-precision tracking and, meanwhile, keep concerned parameters within the boundaries.

For computational efficiency, the constrained FFMPC with TPRRC is formulated into a QP problem to solve for f_1 and f_2

$$\begin{aligned} \min_{\xi} \quad & \xi^T Q^T Q \xi \\ \text{s.t.} \quad & A \xi \preceq h \end{aligned} \quad (3)$$

where ξ , as later defined in (8), is the optimized variable including feedforward inputs, Q is a weighted matrix [see (9)], A linearly models the system singular, and h indicates the constraints. The derivation of the objective [see (7)] and constraints [see (18) and (24)] are presented in the following subsections, where specific expressions of ξ , Q , A , and h are given. Note that the objective is to minimize the tracking error, i.e., $r - y$, subject to the master-slave configured TPRRC feedback systems and the upper/lower limits of physical constraints.

In this study, receding horizon control with the horizon length n is applied, and therefore, this QP problem is solved iteratively. In the receding horizon control, n -step prediction is included in the optimization. After the optimization is solved, control commands are applied to the *simulated* systems, and the horizon will be moved one step forward in the next iteration. Besides, feedforward signals f_1 and f_2 are stored step by step for the later online implementation. To include the system states such as control signals and outputs in the whole horizon, the vector representation of these variables within the moving horizon is introduced as follows:

$$\mathbf{r} \equiv [r(t) \ r(t + \Delta t) \ \dots \ r(t + (n - 1)\Delta t)]^T \quad (4)$$

and similarly for $\mathbf{f}_i$, $\mathbf{u}_i$, $\mathbf{w}_i$, $\mathbf{y}_i$, and $\mathbf{e}_i$. Note that Δt is the control sampling period. As labeled in Fig. 1, $\mathbf{r}$ stands for the reference, $\mathbf{f}_i$ indicates the optimal feedforward signal, $\mathbf{u}_i$ is the control command to the physical plant, $\mathbf{w}_i$ is the difference between the output $\mathbf{y}_i$ and compensation signal generated by the repetitive controller C_{r_i} , and $\mathbf{e}_i$ means the error signal. The subscript $i = 1, 2$ represents the primary-stage and secondary-stage actuators, respectively.

A. Objective Function: Tracking Error Minimization

To optimize the tracking performance within the horizon, the objective function is considered as the weighted quadratic sum of the tracking errors

$$\begin{aligned} & (\mathbf{r} - \mathbf{y})^T W^T W (\mathbf{r} - \mathbf{y}) \\ &= (\mathbf{r} - \mathbf{y}_1 - \mathbf{y}_2)^T W^T W (\mathbf{r} - \mathbf{y}_1 - \mathbf{y}_2) \\ &= (\mathbf{e}_1 - \mathbf{y}_2)^T W^T W (\mathbf{e}_1 - \mathbf{y}_2) = \mathbf{e}_2^T W^T W \mathbf{e}_2. \end{aligned} \quad (5)$$

After exploiting the special structure of the master-slave configuration, the objective function only puts penalty on $\mathbf{e}_2$. The weighting function W is selected to be an identity matrix with n_d look ahead steps

$$W \equiv \text{diag}([\underbrace{0 \ 0 \ \dots \ 0}_{n_d} \ 1 \ 1 \ \dots \ 1]). \quad (6)$$

From (6), only predicted tracking error between the $(n_d + 1)$ th and the n th step is penalized. The number of the look-ahead steps n_d must be smaller than the receding horizon length n , while exceeding the number of unstable zeros in the plant model to prevent implicit unstable pole-zero cancellation. Details of

analysis and discussion of look ahead steps can be found in [34] and [35].

The objective function [see (5)] is then transformed into the typical QP form

$$(\mathbf{r} - \mathbf{y})^T W^T W (\mathbf{r} - \mathbf{y}) = \mathbf{e}_2^T W^T W \mathbf{e}_2 = \xi^T Q^T Q \xi \quad (7)$$

where

$$\xi \equiv [\mathbf{y}_1^T \ \mathbf{y}_2^T \ \mathbf{u}_1^T \ \mathbf{u}_2^T \ \mathbf{f}_1^T \ \mathbf{f}_2^T \ \mathbf{w}_1^T \ \mathbf{w}_2^T \ \mathbf{e}_1^T \ \mathbf{e}_2^T]^T \quad (8)$$

$$Q \equiv W [0 \ \cdots \ 0 \ I]. \quad (9)$$

B. Constraints: Dynamics and Physical Constraints

There are two kinds of constraints in the proposed FFMPC formulation. The first group is equality constraints for the plants' and controllers' dynamics (i.e., the master-slave configured TPRRC feedback system). The second one is inequality constraints defined by the physical constraints of the dual-stage actuators. For the application of MPC, the equality constraints for the system dynamics are always included in its formulation. Nevertheless, realistic constraints are considered by including corresponding inequality constraints in the numerical optimization to protect the actuators from constraint violations.

To describe the system dynamics in receding horizon control, a notation $\mathcal{T}$ is introduced to represent the input-output relation of a plant or controller, also known as the block Toeplitz matrix of the system's Markov parameters [see (11)] and the observability matrix [see (12)]. For example, given any input vector $\mathbf{u}$ and initial states x to an arbitrary discrete dynamic model G , its output vector $\mathbf{y}$ within the horizon can be expressed as

$$\mathbf{y} = \mathcal{T}_{uy} \mathbf{u} + \mathcal{T}_{xy} x \quad (10)$$

$$\mathcal{T}_{uy} = \begin{bmatrix} D_G & 0 & \cdots & 0 \\ C_G B_G & D_G & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ C_G A_G^{n-2} B_G & \cdots & C_G B_G & D_G \end{bmatrix} \quad (11)$$

$$\mathcal{T}_{xy} = \begin{bmatrix} C_G \\ C_G A_G \\ \vdots \\ C_G A_G^{n-1} \end{bmatrix} \quad (12)$$

where (A_G, B_G, C_G, D_G) is the discrete-time state-space expression of G , and $\mathcal{T}_{uy}$ and $\mathcal{T}_{xy}$ relate $\mathbf{y}$, $\mathbf{u}$, and x . Note that this relation can also be found when G is nonlinear.

After mapping the input-output relation by applying (10), the dual-stage system dynamics shown in Fig. 1 can be formulated as equality constraints

$$\mathbf{y}_i = \mathcal{T}_{xy_i} x_i + \mathcal{T}_{uy_i} \mathbf{u}_i \quad (13)$$

$$\mathbf{u}_i = \mathcal{T}_{qu_i} q_i + \mathcal{T}_{wu_i} \mathbf{w}_i + \mathbf{f}_i \quad (14)$$

$$\mathbf{w}_i = \mathcal{T}_{pw_i} p_i + \mathcal{T}_{ew_i} \mathbf{e}_i - \mathbf{y}_i \quad (15)$$

$$\mathbf{e}_1 = \mathbf{r} - \mathbf{y}_1, \quad \mathbf{e}_2 = \mathbf{e}_1 - \mathbf{y}_2 \quad (16)$$

where x_i , q_i , and p_i are initial conditions for G_i , C_i , and C_{r_i} , respectively, $\mathcal{T}_{xy_i}$ and $\mathcal{T}_{uy_i}$ are derived from the plant model G_i ,

$\mathcal{T}_{qu_i}$ and $\mathcal{T}_{wu_i}$ are derived from the prestabilized controller C_i 's dynamics, and $\mathcal{T}_{pw_i}$ and $\mathcal{T}_{ew_i}$ are derived from the dynamics of TPRRC C_{r_i} . Note that i equals 1 for the primary stage and 2 for the secondary actuator.

In the cases where multirate control is required, another group of equality constraints has to be included to describe the zero-order-hold behavior of the lower sampling system (and the horizon n must be integer times of k – the sampling frequency ratio of the primal and secondary stages). For example,

$$T_1 \mathbf{u}_1 = T_2 \mathbf{u}_1 = \dots = T_k \mathbf{u}_1 \quad (17)$$

where T_j indicates the $(j + ck)$ th elements; c equals 0, 1 ... $(n/k) - 1$.

To simplify the notation, we assume that the sampling rate of the primary and secondary stages is the same, i.e., $k = 1$. The equality constraints that describe system dynamics have been further organized as

$$A_d \xi = h_d \quad (18)$$

where

$$A_d =$$

$$\begin{bmatrix} I & 0 & -\mathcal{T}_{uy1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & I & 0 & -\mathcal{T}_{uy2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I & 0 & -I & 0 & -\mathcal{T}_{wu1} & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & -I & 0 & -\mathcal{T}_{wu2} & 0 & 0 \\ I & 0 & 0 & 0 & 0 & 0 & I & 0 & -\mathcal{T}_{ew1} & 0 \\ 0 & I & 0 & 0 & 0 & 0 & 0 & I & 0 & -\mathcal{T}_{ew2} \\ I & 0 & 0 & 0 & 0 & 0 & 0 & 0 & I & 0 \\ 0 & I & 0 & 0 & 0 & 0 & 0 & 0 & -I & I \end{bmatrix} \quad (19)$$

$$h_d = [x_1^T \mathcal{T}_{xy1}^T \quad x_2^T \mathcal{T}_{xy2}^T \quad q_1^T \mathcal{T}_{qu1}^T \quad q_2^T \mathcal{T}_{qu2}^T \quad p_1^T \mathcal{T}_{pw1}^T \quad p_2^T \mathcal{T}_{pw2}^T \quad \mathbf{r}^T \quad \mathbf{0}^T]^T. \quad (20)$$

Most importantly, it is allowable to intuitively incorporate physical constraints of the actuators in the current optimization structure. For example, an inequality constraint can be added, but not restricted, if there is a control saturation for the first stage

$$\underline{u}_1 \leq \mathbf{u}_1 \leq \bar{u}_1 \quad (21)$$

where $\underline{u}_1$ and $\bar{u}_1$ indicates the range of acceptable control inputs. Linear operation is also compatible with linear inequality formulation. For instance, a velocity constraint can be attached to the secondary stage, as shown in (22)

$$v_2 \leq \mathbf{D} \mathbf{y}_2 \leq \bar{v}_2 \quad (22)$$

$$\mathbf{D} = \frac{1}{\Delta t} \begin{bmatrix} -1 & 1 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & -1 & 1 \end{bmatrix}. \quad (23)$$

These two examples that characterize system constraints can be further written as

$$A_c \xi \preceq h_c \quad (24)$$

where

$$A_c = \begin{bmatrix} 0 & 0 & I & 0 & \cdots & 0 \\ 0 & 0 & -I & 0 & \cdots & 0 \\ 0 & \mathbf{D} & 0 & 0 & \cdots & 0 \\ 0 & -\mathbf{D} & 0 & 0 & \cdots & 0 \end{bmatrix} \quad (25)$$

$$h_c = [\bar{u}_1 \mathbf{1}^T - \underline{u}_1 \mathbf{1}^T \bar{v}_2 \mathbf{1}^T - \underline{v}_2 \mathbf{1}^T]^T. \quad (26)$$

Other constraints of higher order dynamics such as acceleration, jerk, or frequency-weighted limits can be similarly included with appropriate filtering matrices in place of $\mathbf{D}$ for velocity constraints. To summarize, the QP formulation for the FFMPC is derived as follows:

$$\begin{aligned} \min_{\xi} \quad & \xi^T Q^T Q \xi \\ \text{s.t.} \quad & A\xi \preceq h \end{aligned} \quad (27)$$

in which

$$A = \begin{bmatrix} A_d \\ -A_d \\ A_c \end{bmatrix}, \quad h = \begin{bmatrix} h_d \\ -h_d \\ h_c \end{bmatrix}$$

where A_d , h_d , A_c , and h_c are defined in (19), (20), (25), and (26), respectively.

The dimension of this QP formulation (i.e., the vector length of ξ) can be further reduced by plugging in the equality constraints of the plant dynamics [see (13)–(16)] into the inequality constraints. By doing so, the dependent variables in (8) are eliminated, and only the independent variables $\mathbf{f}_1$ and $\mathbf{f}_2$ remain. Theoretically, the optimal solutions before and after this reduction are identical. The dimension of the optimization problem is reduced from $10n$ to $2n$, and all the equality constraints are eliminated. This reduction is useful for efficient offline computation. The detail of reduction is omitted here to save space, since it does not affect online implementation and tracking performance. In closing, the constrained FFMPC signals for the dual-stage system can be generated offline by using the following steps.

Generation of the offline FFMPC signals for the dual-stage system

1. Set initial values x_i, q_i, p_i
 2. Set receding horizon length n and look ahead steps n_d
 3. Derive block Toeplitz matrices and observability matrices $\mathcal{T}$
 4. Set $t = 0$
 5. Optimization for the QP problem [see (27)]
 6. Update system states x_i, q_i, p_i by the first elements of $\mathbf{u}_i, \mathbf{w}_i, \mathbf{e}_i$
 7. Save the first element of $\mathbf{f}_i$ as feedforward signal $f_i(t)$
 8. $t: t + \Delta t$, back to step 5
-

C. Effect of Model Uncertainty

Model-based feedforward control is sensitive to model uncertainty; therefore, the FFMPC must be combined with the

robust repetitive control to achieve precise near-periodic trajectory tracking. First of all, robust stability is guaranteed by the TPRRC. Adding feedforward actions to the stable feedback loops does not affect closed-loop stability. The effect of model uncertainty on tracking performance is then analyzed as follows. The dual-stage total tracking error e_2 can be written as

$$\begin{aligned} e_2 &= (r - y_1) - y_2 = r - y \\ &= S_1 S_2 r - \frac{G_1}{1 + C_1 G_1} S_1 S_2 f_1 - \frac{G_2}{1 + C_2 G_2} S_2 f_2. \end{aligned} \quad (28)$$

The first term is contributed by the master–slave configured TPRRC, and the second and third terms are the amount the FFMPC is supposed to improve. Particularly, the fundamental frequency and its harmonics have $\|S_1\|$ and $\|S_2\|$ approximates 0. It turns out that $\|e_2\|$ will still approximate 0 at harmonic frequencies even if the modeling error is not negligible.

Since f_1 and f_2 minimizes the tracking error e_2 in the proposed FFMPC, the bandwidths of f_1 and f_2 are implicitly limited by the reference r . Although the tracking performance of non-periodic components is sensitive to plant uncertainty, at high frequency where modeling error is usually large, the feedforward actions are disabled and, therefore, would not deteriorate the tracking performance.

IV. SIMULATION AND EXPERIMENTAL RESULTS ON A MULTIRATE DUAL-STAGE POSITIONING SYSTEM

A. Systems, Models, and Constraints

The proposed fusion of FFMPC and TPRRC was demonstrated on a dual-stage fast tool servo system, consisting of an LM [36] as the primary stage and a PZT [37] as the secondary stage, as shown in Fig. 3. This system was designed for high-precision noncircular machining such as engine pistons. The bode plots of each open-loop plant model G_i are shown in Fig. 4, which were obtained by the sinusoidal frequency sweep method. The LM G_1 was modeled as a sixth-order type 1 system with dominant poles at 0, -210.5 , and -230.4 ; the PZT G_2 was modeled as a fourth-order system with a resonant peak at 5 kHz. The bandwidth of the motor is around 100 Hz, which is much narrower than the PZT.

The open-loop plants G_1 and G_2 were first closed by C_1 and C_2 with the structure shown in Fig. 1, where C_1 was selected as a proportional-derivative (PD) controller and C_2 was a proportional-integral (PI) controller. Due to the different actuator bandwidth, the LM is closed at 10-kHz sampling rate and the PZT is closed at 20-kHz sampling rate.

The TPRRC design satisfies the weighted sensitivity $|W_p S| < 1$ (see Fig. 5) and robust stability $|W_r T| < 1$ (see Fig. 6) for each of the two plant models. For the FFMPC, the design constraints included in the design are listed in Table I. Though each actuator has its own travel range, the LM has a travel range much larger than the tooling profile, hence is not concerned. Besides, the actuator amplifiers are limited by the electric current and voltage capacity. The motor amplifier current command relates to the motor force or acceleration, and the voltage relates to the motor velocity. The PZT's amplifier

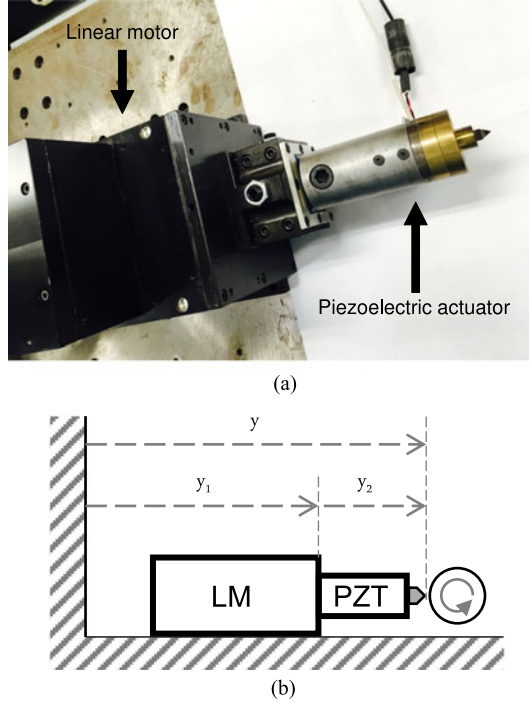


Fig. 3. Dual-stage fast tool servo system. (a) Experimental platform. (b) Schematic diagram.

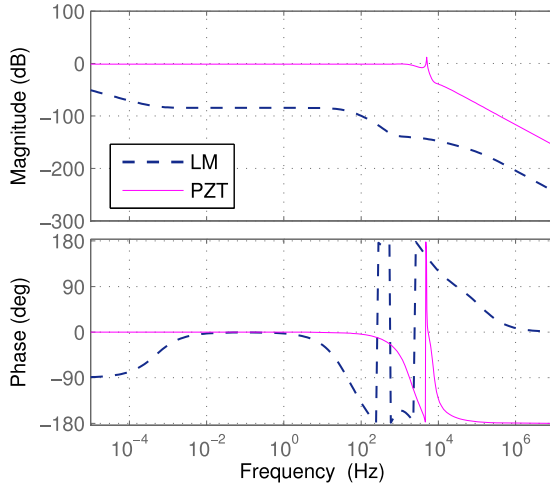


Fig. 4. Bode plots of the continuous-time open-loop plant models.

voltage command relates to the PZT's displacement, and the current relates to the velocity. Also, the velocity of the LM must be limited to ensure that the optical encoder operates robustly. In the actual hardware, the physical limits are 10–30% larger than these design constraints to account for unmodeled dynamics and disturbances.

In the following simulation and experiments, the performance of the proposed FFMPC and TPRRC is compared to the results when only the TPRRC feedback controllers were applied. The simulation will validate the algorithm with respect to the plant model, design constraints, and reference trajectory, while the experiment will demonstrate the actual performance achieved subject to the experimental system's uncertainties.

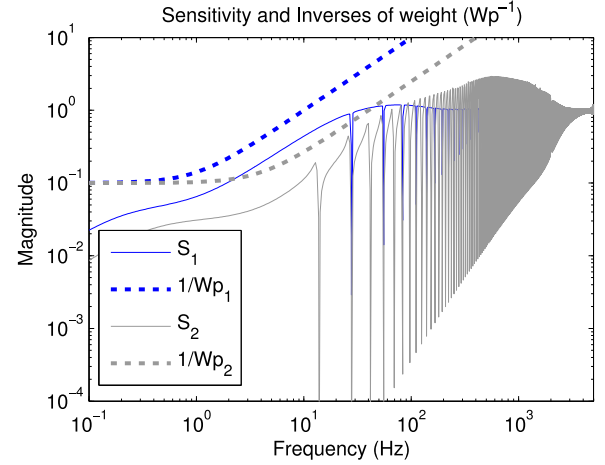


Fig. 5. Sensitivity analysis of the TPRRC.

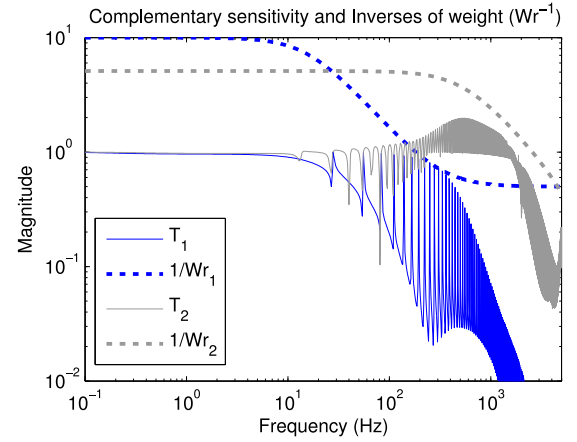


Fig. 6. Complementary sensitivity analysis of the TPRRC.

TABLE I
DESIGN CONSTRAINTS OF THE DUAL-STAGE SYSTEM

Limitation	PZT position [μm]	LM current command [A]	PZT command [V]	PZT current [A]	LM velocity [m/s]
Upper	12	25	7.3	0.1	0.03
Lower	−12	−25	0	−0.1	−0.03

B. Simulation Results

A challenging engine piston profile is selected to demonstrate the performance of the proposed approach. The sampled trajectory (at the feed rate of 0.167 mm/rev) is shown in Fig. 7, which is composed of periodic oscillations (e.g., between 5.0 and 5.3 s, as shown in the zoomed-in section), sharp turns, and drastic ramping (between 7 and 8.5 s). The constrained FFMPC signals were generated for this specific reference after solving the proposed QP optimization offline and then were directly fed to the actuators G_i . Note that the CVX solver in MATLAB [38] was applied to solve the QP problem [see (27)].

As shown in the previous sections, the TPRRC was applied to achieve the asymptotically tracking performance of near-periodic components in the desired trajectory. Since there are mirrored patterns in the tracking reference (see Fig. 7, bottom

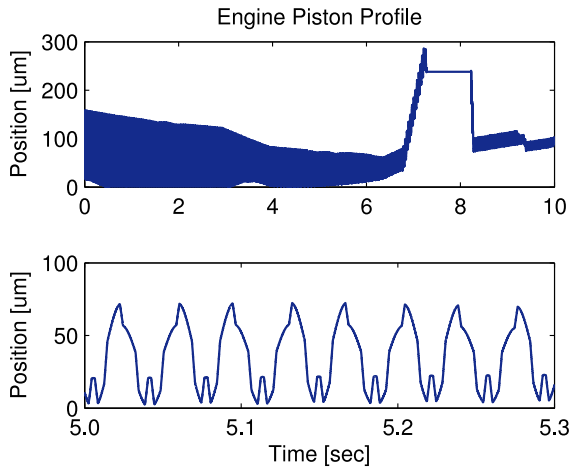


Fig. 7. Reference trajectory for machining the engine piston.

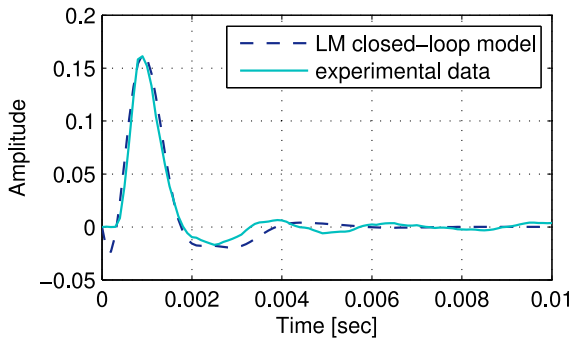


Fig. 8. Normalized impulse response of the closed-loop LM.

figure), the harmonics to be removed by the primary-stage repetitive control was set to be 27 Hz and 13.5 Hz for the secondary stage. As a result, C_{r1} was the 740th order and C_{r2} was the 1450th order. Although the orders of the repetitive controllers were very high, most of the states were just shift registers for time delays. Therefore, TPRRC is online implementable, but challenges online MPC implementation due to its numerical high order.

As for the design of offline FFMPC, the entire closed-loop system with G_i , C_i , and C_{r_i} was considered, as discussed in Section III. The total receding horizon length n was selected as 2 ms (40 steps under 20-kHz sampling) in order to cover the system dynamics of the lower bandwidth LM, as shown in Fig. 8. The main peak of the impulse response of the LM was included within the first 2-ms horizon. After 2 ms, the modeled dynamics deviate from the experimental impulse response. This suggests that even if the horizon was larger than 2 ms, the performance improvement of online implementation would still be limited because of the modeling error. Besides, the look ahead steps n_d was set to be 4 based on the number of nonminimum phase zeros in the plant model of the LM.

The *simulated* performance after combining FFMPC and TPRRC was first examined. The results in Fig. 11 show that adding the FFMPC feedforward signal to the TPRRC feedback control loop significantly improved the tracking performance on both the LM (upper figure) and the total tracking error

(lower figure). The initial slow error convergence in the repetitive control was eliminated. The zoomed-in section where the challenging near-periodic piston profile is signified (see Fig. 12) shows that even without the factors of the constraint violations or initial transient convergence, the feedforward action reduced the tracking error of the repetitive control feedback loop. Of course, FFMPC alone without TPRRC would not perform well, since the model-based optimization is sensitive to uncertainties like modeling errors and disturbances.

Adding FFMPC also eliminated constraint violations and thus also contributed to improving the tracking performance. The monitored physical variables (see Table I) are shown in Fig. 9, where the limits are also illustrated by the dash lines. Without injecting the FFMPC signals, the piezoelectric amplifier's command almost saturated at all times and the PZT displacement was saturated frequently. On the other hand, the LM current command and piezoelectric amplifier's current were mostly not even close to their limits. Contrarily, the addition of the constrained FFMPC made all parameters staying in the acceptable range. Note that the LM command and piezoelectric amplifier's current were significantly increased and closer to their limits, but they never exceeded the design boundaries.

C. Experimental Results

The same reference and the corresponding FFMPC signals were then applied to the actual system. To enable fast sampling rates, a dual-core (Intel Core i3-2120 CPU @ 3.3 GHz) desktop PC equipped with a National Instruments R-series field programmable gate array (FPGA) board (PCIe-7841R) was serving as the real-time control processor in the experiments. An optical encoder with 0.1- μm resolution and an analog capacitive sensor with sensitivity 0.80 V/ μm were used to measure the output positions of the LM and the PZT, respectively. The measurements were sampled, and the commands controlling the LM and the PZT were computed by two independent control loops at 10 and 20 kHz, respectively. In addition to the hardware setup, the proposed synergistic method enables fast-sampling control by applying offline computed feedforward signals. Compared to the dual-stage TPRRC [3] real-time implementation, the FFMPC feedforward signals stored in a lookup table are added to the TPRRC without additional real-time computation.

The experimental results generally show similar characteristics as those in the simulation, while the differences are due to plant model uncertainties and the experimental system's sensor resolution and noise level. The addition of the FFMPC to the TPRRC improved the tracking performance and reduced the constraint violation rate. With the TPRRC only, the constraint violation rate was 12.7% over the whole tracking trajectory (see Fig. 10), and most of them were contributed by piezoelectric amplifier's voltage command saturation due to the slow transient response by the primary LM stage. Adding the FFMPC reduced the constraint violation rate to 0.60%. Ideally, the violation rate should be zero. Nevertheless, due to modeling error (see Fig. 8), constraints were sometimes violated during the experiment. The actual hardware limitations were not violated, since some margins of the limits had been reserved to protect the hardware

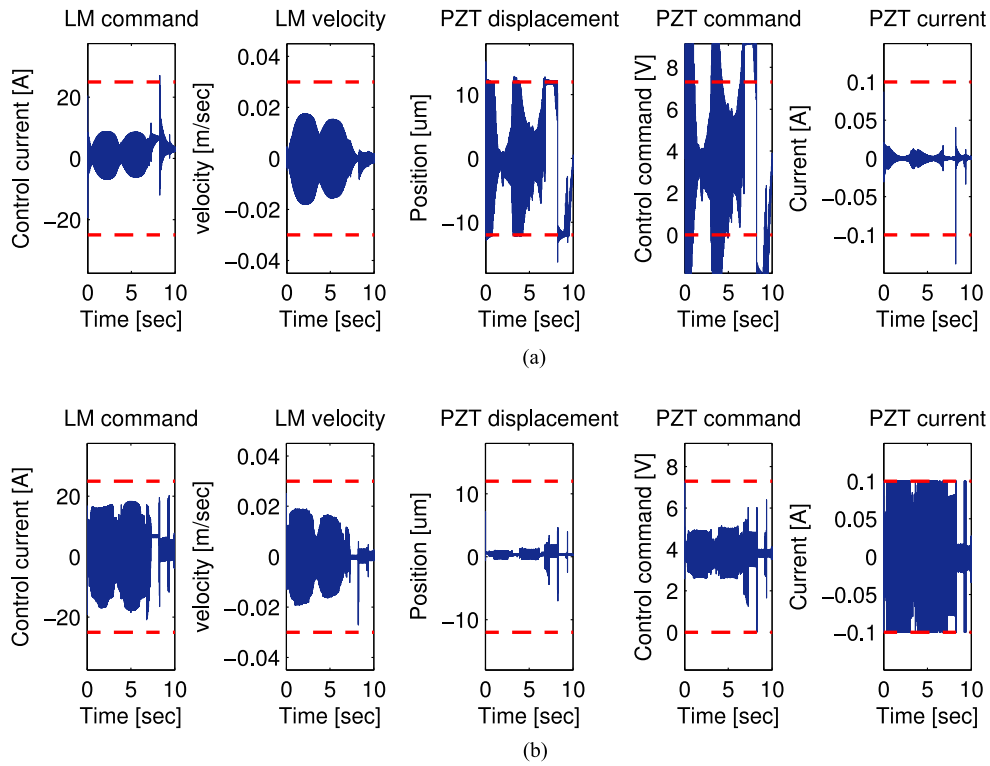


Fig. 9. Simulation results of constraint violations with only feedback (TPRRC) and after adding feedforward (FFMPC). Red-dash lines represent the design constraints incorporated in the optimization (see Table I). (a) Only TPRRC. (b) Combining FFMPC and TPRRC.

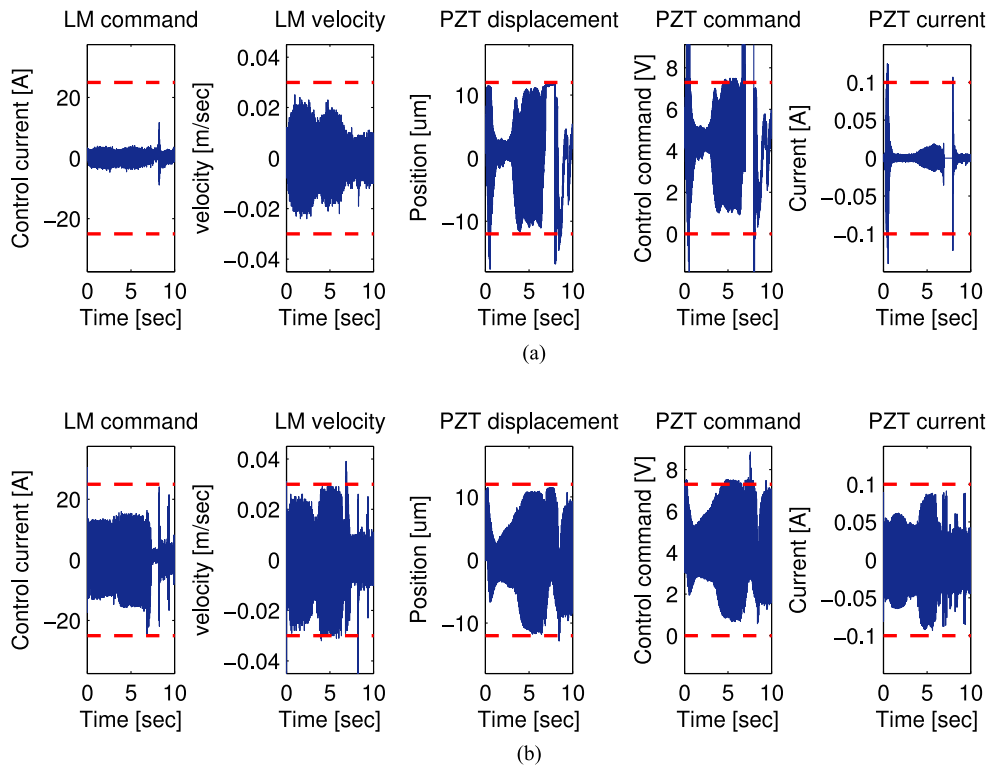


Fig. 10. Experimental results of constraint violations with only feedback (TPRRC) and after adding feedforward (FFMPC). Red-dash lines represent the design constraints incorporated in the optimization (see Table I). (a) Only TPRRC. (b) Combining FFMPC and TPRRC.

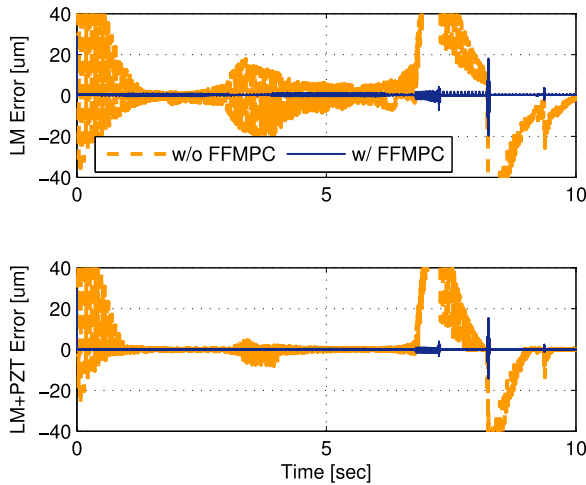


Fig. 11. Simulation result of dual-stage tracking—the LM (upper) and the total tracking error (lower) before and after applying feedforward (FFMPC).

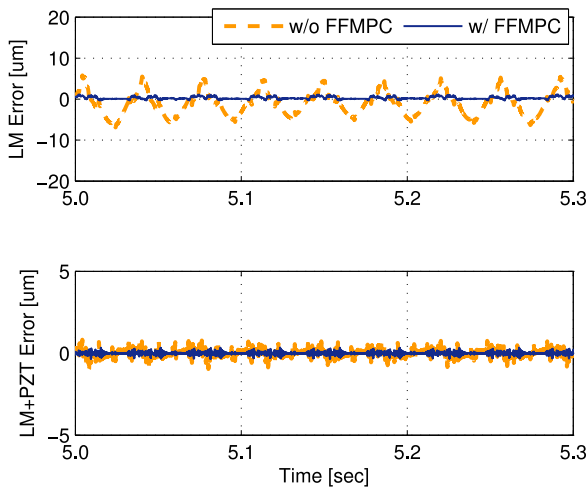


Fig. 12. Simulation result of dual-stage tracking at the periodic section of the trajectory.

TABLE II
EXPERIMENTAL CONSTRAINT VIOLATION RATES

Violation Rates	PZT position	PZT command	PZT current	LM velocity	Overall
only TPRRC	0.77%	12.08%	0.02%	0.00%	12.70%
FFMPC+ TPRRC	0.35%	0.25%	0.01%	0.02%	0.60%

LM current commands were never saturated.

from actual constraint violations. The signal levels of the LM command, velocity, and piezoelectric amplifier's current were increased with the addition of the FFMPC as the simulation had predicted, but they still did not exceed the design constraints in the experiment with the presence of modeling error. Individual constraint violation rates are listed in Table II.

Adding the FFMPC feedforward signal to the TPRRC feedback control loop reduced the root mean square (RMS) value of the total tracking error from 12.22 to 1.86 μm . Fig. 13 shows that the initial slow error convergence in the repetitive control,

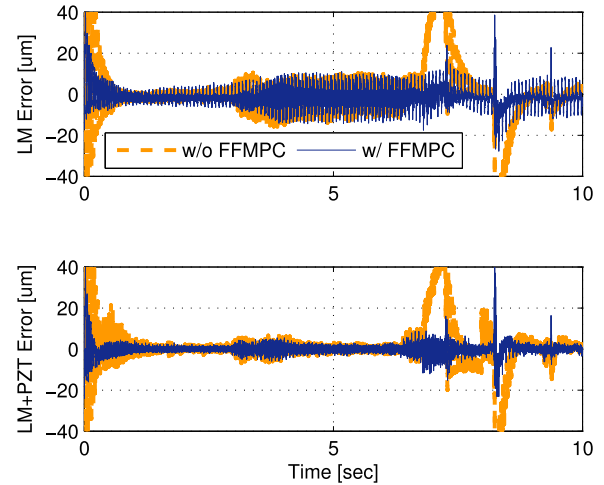


Fig. 13. Experimental result of dual-stage tracking—the LM (upper) and the total tracking error (lower) before and after applying feedforward (FFMPC).

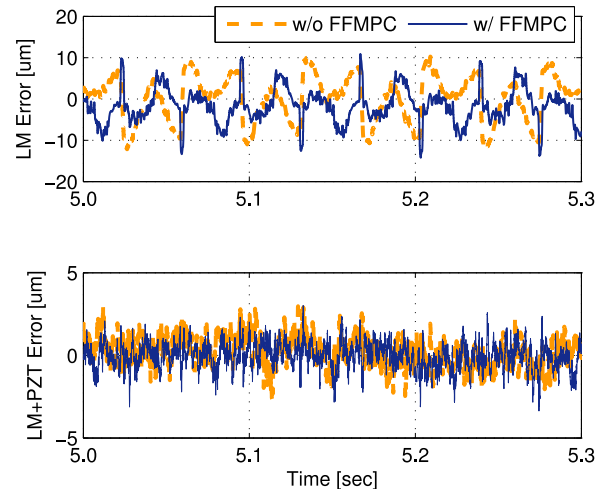


Fig. 14. Experimental result of dual-stage tracking at the periodic section of the trajectory.

though not eliminated like in the simulation, was significantly improved with the FFMPC. Adding the FFMPC also eliminated the constraint violations and thus contributed to improving the tracking performance. The drastic ramping during 7–8.5 s caused large errors in the LM control and saturation of the PZT. By improving the tracking performance of the LM (upper figure in Fig. 13), the feedforward action eliminated the saturations and therefore significantly reduced the total tracking errors (lower figure).

In the zoomed-in section where the repetitive control had reached near-steady-state performance (see Fig. 14), the added feedforward action reduced the tracking error of the repetitive control feedback loop slightly by 16.5% in contrast to the simulation result. This shows that the FFMPC, given its sensitivity of modeling errors, is most beneficial to the repetitive feedback control loop by avoiding constraint violations and improving the transient convergence rate. The spectra of the experimental tracking errors shown in Fig. 15 suggest that the improvement of adding the FFMPC to TPRRC is evident in low frequencies

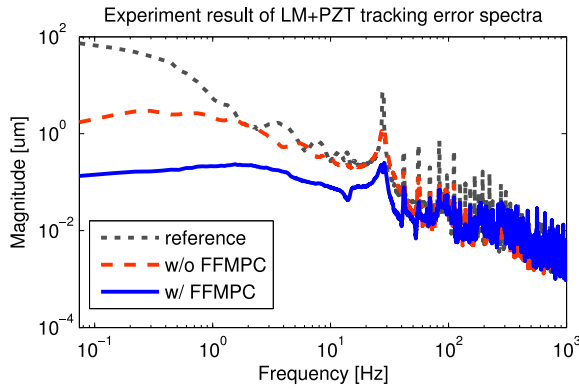


Fig. 15. Comparison of the spectra of tracking error.

TABLE III
EXPERIMENTAL COMPARISONS OF RMS ERROR FOR DIFFERENT
DUAL-STAGE APPROACHES

Stage/Method	Parallel TPRRC	MS config. TPRRC	MS config. TPRRC +FFMPC
LM	14.66	14.73	4.30
LM+PZT	14.68	12.22	1.86

Units are in micrometers.

TABLE IV
EXPERIMENTAL COMPARISONS OF RMS ERROR FOR DIFFERENT
DUAL-STAGE APPROACHES AT THE PERIODIC SECTION OF THE TRAJECTORY

Stage/Method	Parallel TPRRC	MS config. TPRRC	MS config. TPRRC +FFMPC
LM	5.61	5.64	4.52
LM+PZT	5.59	0.91	0.76

Units are in micrometers.

up to near 100 Hz. This can be expected from the LM modeling error where the uncertainty bound W_r (see Fig. 6) crosses 0 dB also near 100 Hz. Finally, the parallel scheme with the reference signal divided into low-pass filtered section for the LM and the high-pass filtered section for the PZT was also implemented for comparison with the master–slave configuration. The results shown in Tables III and IV suggest that the parallel scheme is not as effective as the master–slave configuration.

V. CONCLUSION

We have proposed a method to achieve high-precision tracking of dynamic trajectories for dual-stage actuators and demonstrated its effectiveness. We have motivated and justified the use of the large-stroke low-bandwidth LM and the short-stroke high-bandwidth AZT for tracking complex engine piston profiles in the noncircular turning process. It has also been shown that either the linear low-pass and high-pass filtering or the master–slave approach, to assign the reference trajectory for each actuator, is inadequate in achieving desired performance due to constraint violations.

The offline receding horizon constrained optimization (FFMPC) is effective in determining the optimal feedforward

inputs subject to the realistic physical constraints of each actuator. With inevitable uncertainties in modeling errors and external disturbances, the feedback loop for each actuator is critical to achieving desired tracking performance. Particularly, the two-parameter robust repetitive control is suited for tracking near-periodic trajectories, such as those in tracking the engine piston profiles. The synergy of the FFMPC and TPRRC has been demonstrated by the high-sampling-multirate digital control experimental results, where the RMS tracking error was 1.86 μm , reduced by 84.8% when compared with 12.22- μm error of the TPRRC alone, contributed by the optimization that optimally coordinates the two actuators and eliminates the constraint violations.

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